**Motivation**

- With their nutrition labels, the FDA includes only a small number of nutrients per food (~ 10-15)
    - The reality is that food items are composed of hundreds of biomolecules + there are other ways to categorize or label them that are not necessarily biologically driven
        - While the main goal of the FDA is to protect consumers and help them make healthy decisions, there is a question of whether the current representation of foods does this as effectively as it could

**Models**

- **Overarching Principle**
    - Choose a health outcome, predict the health outcome using a person's demographic features and a representation of diet
        - Identify the representation of diet that leads to greatest predictive success through CV

- **Thematic Models**
    - **Baseline Model**
        - Attempt to predict a health status using
            - Demographic features, such as a person's age, race, income, education, etc.
                - AKA: Controls
            - This model will help to isolate the predictive additions that food representations provide

    - **FDA Model**
        - Attempt to predict health status using
            - Controls
            - Diet, as represented by the FDA's nutrition label

    - **Feature-Selected Model**
        - Perform feature selection (possibly through Lasso) to isolate approximately 10-15 predictors to represent diet
        - Attempt to predict health status using
            - Controls
            - Diet, as represented by the feature selected

    - **Biologically Reasoned Model (OPTIONAL)**

        - Using domain knowledge, isolate and justify 10-15 predictors to represent diet
        - Attempt to predict health status using
            - Controls
            - Diet, as represented by the reasoned predictors

**Data**

- analytic_cohort.csv → 3,286 adults and 305 columns
    - Each row is an adult and contains their demographic information (age, sex, race, income, etc.), health outcomes (BMI, blood pressure, cholesterol, H1ABC, etc.), conventional nutrition intakes averaged over two days, and alternative nutrition intakes averaged over two days